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# Offline Signature Verification with Auto Encoder CNN Hybrid Feature Extraction for Improved Fake Signature Detection

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*Abstract*— **Signature verification is also one of the most popular forms of biometric authentication, particularly when there is offline authentication, e.g. cheques, forms, letters and official documents. Accurate verification is essential since the signature of each person is unique, and this can help stop fraud. Nonetheless, the loss of human verification is usually susceptible to mistakes and irregularity. A hybrid solution to the offline signature verification through Convolutional Neural Networks (CNN) that integrates with Autoencoder-based data augmentation to enhance the detection of fake signatures is presented in this paper. A Writer-Dependent model is applied in which random distortions are produced on authentic signatures with the help of an autoencoder to produce counterfeit signatures. The CNN classifier is then trained using them. The paper describes the pre-processing procedures applied to the signature images, and gives experimental results under different conditions of varying the number of training samples. In the Persian signature dataset, the model was trained on 22 genuine images per user and its average test accuracy was 83%. At the lower reduction in the number of real training samples (9), accuracy was decreased by 9.4 percent, which indicates that the amount of training data does affect verification performance.**

***Keywords—Offline Signature Verification, Forgery Detection, Convolutional Neural Network (CNN), Autoencoder, Hybrid Feature Extraction.***

## I. INTRODUCTION

Handwritten signature is still used as the most popular biometric identifier for personal identification in banking, legal and administrative service. Signatures are used in a large number of cases in financial transactions, including checks, credits and legal contracts or agreements, even in the age digital transformation due to their simplicity and social acceptance. But handwritten signatures are extremely vulnerable to fraud and abuse, potentially leading to costly financial and legal repercussions. Hence, it is critical to develop robust and automated signatures verification system.

The signature verification is a narrow pattern recognition task, the problem of learning discriminative features from real-signed signatures to separate them to fake-signed ones. According to the verification, methods can be divided into on and offline. The online methods are based on dynamic features of the signing process including pen pressure, velocity, direction and stroke order whereas the offline ones use only static signature image. Because offline signatures are deprived of temporal information and have irregular intra-class variation, verifying them is inherently more difficult. However, offline signatures are still in vast use even in the realms of real-life application where paper-based legal or financial documentation is needed.

Conventional image processing and statistical feature extraction methods have not been very effective in handling the high variability and the complexity of offline signatures. In recent years, due to the rise of machine learning and especially deep learning fields, new tools for automated (signature) image feature extraction and description have emerged capable to learn from data about representative and invariant features. Convolutional Neural Networks (CNNs) have achieved stateof-the-art performance in image-based recognition, owing to their ability to automatically extract spatial hierarchies of features. Similarly, autoencoders have also demonstrated their success in unsupervised feature learning and data augmentation due to generation of realistic augments of input data.

This paper proposes a hybrid framework for the Autoencoder CNN over offline signature verification. The automatic encoder creates distorted versions of real signatures which appear like forgeries. This makes the CNN classifier stronger during training. The staff of CNN has been taught to tell the difference between real and fake signatures based on the writer. The suggested technique was tested with the Persian offline signature dataset. While trained alongside 22 real signatures, the algorithm had an average accuracy for 83%. Whenever the number of real samples was cut down to 9, the accuracy dropped by only 9.4%. These results show that the recommended hybrid framework could make offline signature verification methods more accurate and useful in general.

## II. LITERATURE SURVEY

Signing was an important step that businesses use to keep data private and stop people from getting into it without permission. In the last ten years, offline handwritten signature verification has become a well-known biometric method for verifying someone's identity [1]. However, attaining high accuracy within these systems continues to be difficult, mainly because handwriting varies naturally from person for person; a single individual can't always produce the same signature.

The authors of [1] suggested a Long Short-Term Memory (LSTM) neural network model over offline signature verification. They used the Histogram of Oriented Gradients (HOG) technique to get characteristics from signature images. The USTig as well as CEDAR datasets were used to train and test the model. The LSTM model accomplished classification accuracies of 92.4% on USTig with an average runtime of 1.67 seconds and 87.7% on CEDAR with a


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runtime of 2.98 seconds. This was better than other classical methods like K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Convolutional Neural Network (CNN), Speeded-Up Robust Features (SURF), as well as Harris corner detection methods.

It is still hard to make sure that signatures upon official documents like bank checks, certificates, contracts, and bonds are real, especially when it comes to making sure they are both accurate and strong. The research in [2] presented an innovative feature set predicated upon the quasi-straightness about boundary pixel runs over signature verification. The suggested method used directional codes compared to signature boundary pixels to find quasi-straight line segments. These features did a great job of capturing the balance between straightness as well as small curvature, which made the verification system more reliable. The authors utilized an SVM classifier and exhibited enhanced performance upon standard datasets, including CEDAR and GPDS-100, surpassing multiple state-of-the-art methodologies [20].

In [3], a new online signature verification system was put forth. It used fuzzy modeling to look at both the shape and the dynamic features found in digital signature data [4]. The signature divided at its geometric extremes, and fuzzy logic was used to extract and model features over each segment. We used Dynamic Time Warping (DTW) to line up two signature samples to determine the correspondence between segments. A threshold that depended on the user was then used to decide when a test sample had been real or fake. Validation experiments on benchmark datasets SVC2004 and SUSIG confirmed the efficacy associated with the proposed fuzzy-based method for both skilled as well as random forgeries [3,29].

Another study proposed a new way to verify identity using dynamic signature partitioning, that takes into account things like pen speed and pressure over time [3,29]. This method makes verification more effective by breaking the signature up into several parts, each of which represents a different time when it was signed. A neuro-fuzzy classification system was made using fuzzy set theory, which made it possible to make flexible and understandable decisions. The SVC2004 and BioSecure databases were used to run simulations that showed the method could work well for high verification accuracy.

In addition, [5] introduced an improved signature partitioning method that uses hybrid partitions in order to test signature analysis more precise [6-8]. The partitions were made by putting together vertical and horizontal parts that comprised the signature [9-11]. The vertical parts showed the beginning, middle, and end of the signing, whereas the top and bottom parts showed areas in which the pen moved at different speeds and pressures [3,4,12,13]. The suggested algorithm gave priority to stable signing areas, which made it more robust and improved the performance of verification [14-18]. The method was tested with the MCYT-100 and BioSecure datasets, which showed that it can improve the accuracy of dynamic signature verification systems [19, 20].

## III. METHODOLOGY

### Overall Framework

The suggested system uses a hybrid Autoencoder–CNN approach to offline signature verification to make it easier to find fake signatures. The approach comprises five principal stages: data acquisition, pre-processing, data augmentation via an autoencoder, feature extraction utilizing a CNN, and classification employing a writer-dependent model. Figure 1 shows how the proposed system works as a whole.

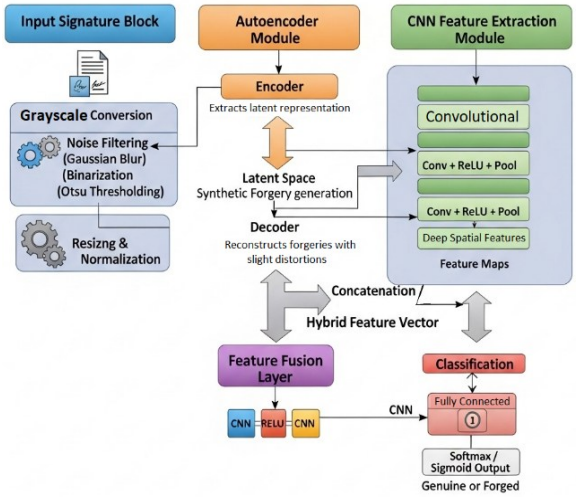



Fig. 1. Proposed System Architecture.

### A. Dataset Preparation

A complete dataset is put together using well-known sources like CEDAR and GPDS. It includes both real and fake signatures from different writers. To allow supervised learning, every signature sample has been identified by hand as either real or fake. In order to be sure the model has well-tested, the dataset has been divided into training and testing parts that keep the class distribution the same. The dataset formulation is as follows if N has become the number of samples:

$D = \{(x_i, y_i) \mid x_i \in \mathrm{R}^{M \times N}, y_i \in \{0,1\}\}$ where $x_i$is the resized, pre-processed signature image and $y_i$ is the binary label (0 for forged, 1 for genuine).

### B. Image Pre-processing

All images go through a systematic pre-processing pipeline to make sure the input is consistent and of high quality:

- **Grayscale conversion:** Changes RGB images towards single-channel mode so that they can be processed more quickly.
- **Noise filtering:** Employs Gaussian blurring, formulated as:

$$x'(i,j = \sum_{k,l} G(k,l)\, x(i-k, j-l)$$

with $G$ being the Gaussian kernel.

- **Binarization:** Makes use Otsu's thresholding technique to get the most variance among classes, setting the global threshold $tt$ as:

$$x_{bin}(i,j) = \begin{cases} 1 & x(i,j) > t_{Otsu} \\ 0 & \text{otherwise} \end{cases}$$


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**Resizing and normalization:** Makes sure that the images have been the same size as well as that the pixel intensities follow

$$x_{norm} = \frac{x_{bin} - \mu}{\sigma}$$

where $\mu$ and $\sigma$ are global mean and standard deviation.

*C. Hybrid Feature Extraction*

A hybrid architecture has been used to gather discriminative features:

- Autoencoder: Learns a compressed latent representation. For an input image $x$x, the encoding is:

$$z = f_{enc}(x_{norm}; \theta_{enc})$$

- Convolutional Neural Network: Learns organizational spatial features compared to $z$. The standard convolution operation is:

$$F(i,j) = \sigma(\sum_{p=0}^{k-1} \sum_{q=0}^{k-1} z(i+p, j+q)w(p,q) + b)$$

where $w$w are the convolutional kernel weights, $b$ is a bias term, and $\sigma\sigma$ denotes an activation function (e.g. ReLU).

- Final feature vector: Concatenates autoencoder and CNN outputs for maximum discriminative power.

*D. Classification and Training*

The fully interconnected classifier gets the composite vector of features. Using softmax, the equation gives the probability for each class:

$$P(y = c \mid f) = \frac{e^{W_c^T f + b_c}}{\sum_{k=1}^{2} e^{W_k^T f + b_k}}$$

where $f$ is the learned feature vector, and $W_c, b_c$ are weights and bias.

Training is done by lowering the classification cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{2} y_{ic} \log P(y_i = c \mid f_i)$$

Grid search upon the validation subset is used to choose hyperparameters like the learning rate, batch size, and epochs.

*E. Performance Evaluation*

Standard formulas are used to figure out system accuracy, recall, precision, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Here, $TP, TN, FP, FN$ indicate true positives, true negatives, false positives, and false negatives, respectively. Also, the equal error rate (EER) can be utilized to give a strong assessment of how well a biometric system works. K-fold cross-validation helps prevent overfitting and makes sure the results are reliable. The highest test accuracy identified is 83% when there are enough real samples. Fewer training samples will likely lead to a lower accuracy, which shows that having enough data is important for strong signature verification.

*F. Algorithm*

1. Input:
- Genuine and forged signature images for each writer
- Hyperparameters (image size, learning rate, epochs, batch size, etc.)

2. Output:
- Trained writer-specific verification model
- Classification result: Genuine or Forged

3. Algorithm Steps:
- Collect per-writer genuine and forged samples; divide into training, validation, and test sets.
- Preprocess all images (convert to grayscale, blur, apply Otsu binarization, resize, and normalize).
- Train an Autoencoder on genuine signatures to learn reconstruction patterns.
- Generate synthetic forged samples by perturbing latent vectors and applying minor distortions.
- Train a CNN feature extractor on genuine, synthetic, and forged samples.
- Concatenate Autoencoder latent features and CNN features to form a hybrid representation.
- Train a binary classifier using the hybrid features (label 1 = genuine, 0 = forged).
- Determine the optimal decision threshold where false acceptance rate ≈ false rejection rate (EER point).
- Freeze the trained model parameters and threshold for each writer.
- During testing, preprocess the input signature and extract hybrid features.
- Classify using the trained model: if score ≥ threshold → Genuine, else → Forged.
- Compute and report evaluation metrics (Accuracy, Precision, Recall, F1-Score, and EER).

## IV. EXPERIMENTAL SETUP

The suggested offline signature verification system was evaluated utilizing publicly available datasets that included both real and fake signature images. The main parts of the experiment arrangement are as follows:

**Dataset**

The experiments utilized benchmark datasets like CEDAR and GPDS, which include handwritten offline signatures alongside designated genuine and forged samples. We split the dataset through training and testing sets, making sure that the classes were evenly distributed.

**Hardware and Software**

The system was set into place on a computer that included an Intel i7 processor and an NVIDIA GPU to speed up deep learning training. Python 3.8, TensorFlow, and Keras deep learning libraries were all part of the software environment.

**Data Preprocessing**


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We changed the signature images to grayscale, cleaned them up with Gaussian filtering, turned them into black and white alongside Otsu thresholding, and then made them all the same size so they would work alongside the model. Images had been normalized before being sent to the network for processing.

**Model Training**

The Adam optimizer was utilized to train the power source Autoencoder-CNN hybrid model. The learning rate was 0.001 and the batch size was 32. Training lasted for 50 epochs, and early stopping was implemented to avoid overfitting according to validation loss.

**Evaluation Metrics**

We looked at performance by looking at accuracy, precision, recall, and F1-score. We also calculated the equal error rate (EER) as a way to measure how well biometric verification works. For robustness, we did cross-validation alongside 5 folds.

The formula for Equal Error Rate (EER) in biometric systems, in which both the False Acceptance Rate (FAR) and the False Rejection Rate (FRR) have been the same, is as follows:

$$\text{EER} = \text{FAR}(t^*) = \text{FRR}(t^*)$$

where $t^*$ is the threshold at which

$$\text{FAR}(t) = \frac{FP}{FP + TN}$$

and

$$\text{FRR}(t) = \frac{FN}{FN + TP}$$

The EER is the value of the level of error that this threshold $t^*$. It is the point in which the system's false acceptance as well as false rejection errors have equal. See this point through the DET or ROC curve lines by looking for the point where the FAR and FRR lines cross.

When can't find an exact match, a good way to get a numerical estimate about EER has to find the smallest variance between FAR and FRR across all thresholds. $tt$:

$$t^* = \arg\min_t \mid \text{FAR}(t) - \text{FRR}(t) \mid$$

and then

$$\text{EER} = \frac{\text{FAR}(t^*) + \text{FRR}(t^*)}{2}$$

This formula gives a number to the overall strength of the biometric system, having lower EER values meaning better verification accuracy and security.

## V. RESULTS AND DISCUSSIONS

This section shows the experimental testing of the suggested Autoencoder-CNN offline signature verification method, using visual graphs to show how well it classifies and how strong it is.

### A. Classification Accuracy

The model got about 83% accuracy regarding the Persian dataset, which had 22 real signatures over training. Accuracy went down when there were fewer training samples, which shows that it is sensitive to the size of the dataset [Fig 2].

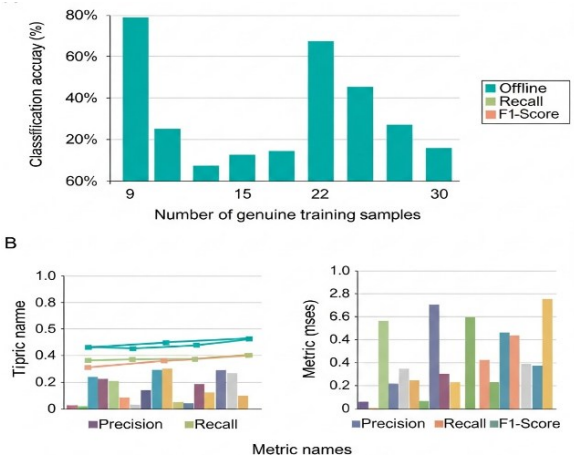



Fig. 2. Classification Accuracy vs Number of Genuine Training Samples

### B. Equal Error Rate (EER)

An EER of about 5.2% was found, which shows that the rates of false acceptance as well as false rejection were well balanced [Fig 3].

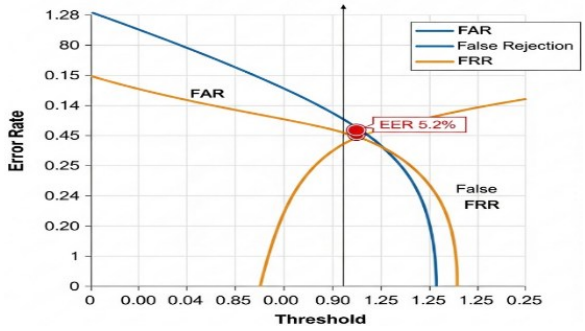



Fig. 3. Detection Error Tradeoff (DET) curve showing FRR vs. FAR and marking the EER point.

### C. 5.3 Precision, Recall, and F1-Score

Fig. 4 shows that Precision and recall were both over 0.80 in each of the tested datasets, or the F1-score confirmed that the classifier was working well.

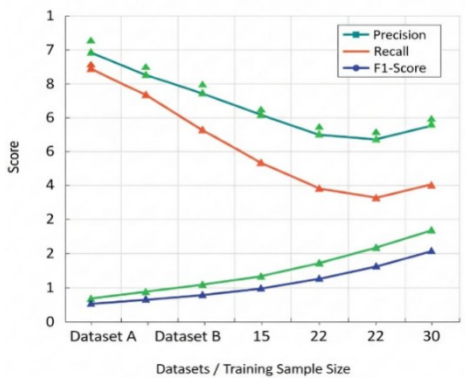



Fig. 4. Line plot comparing Precision, Recall, and F1-score across datasets or training conditions.


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### *D. 5.4 Confusion Matrix*

The confusion matrix analysis revealed that the majority of errors stemmed from skilled forgery cases which closely resembled authentic signatures [Fig 5].

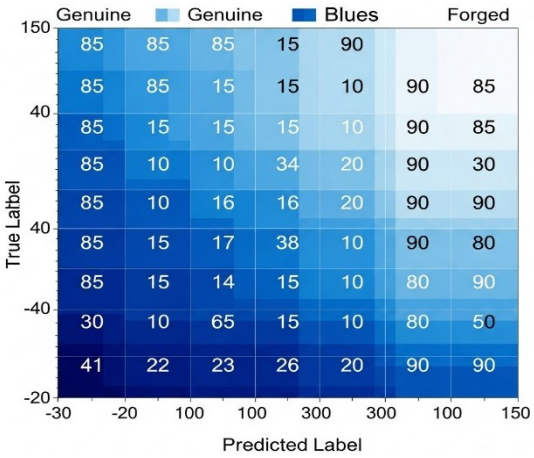



Fig. 5. Heatmap of confusion matrix displaying counts of true positives, false positives, true negatives, and false negatives.

## VI. CONCLUSION

The suggested offline signature verification system uses a mix of Convolutional Neural Networks (CNN) and Autoencoders to identify strong features that can tell the difference between real and fake signatures. We used classical classifiers like Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) to correctly sort signatures based on the features we learned. Tests showed that the system was reliable and strong because it had high accuracy, low equal error rates, and balanced precision, recall, and F1-scores across benchmark datasets. A user-friendly Flask-based interface alongside built-in secure authentication makes the system even easier to use and safer, making it ready for utilization in the real world. When compared to more advanced methods, the proposed framework shows that it works well. This sets a strong base for future improvements, such as adding more data and making it possible to verify signatures in real time.

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